

EDUCATION

National Tsing Hua University

M.S. in Statistics and Data Science, GPA: 4.00/4.30

Hsinchu, Taiwan

2023–2025

- Thesis: “Nonparametric Mediation Analysis of Non-Markov Illness-Death Model”
- Advisor: Prof. Yen-Tsung Huang

National Chung Hsing University

B.S. in Applied Mathematics; Double Major in Finance, GPA: 4.10/4.30

Taichung, Taiwan

2018–2023

RESEARCH INTERESTS

Causal mediation analysis; nonparametric methods; machine learning theory; survival analysis.

EXPERIENCE

Academia Sinica

Research Assistant

Taipei, Taiwan

2025–Present

- Supervisor: Prof. Yen-Tsung Huang
- Project: Multivariate Mediation Analysis via SIR-Based Single-Index Models

Developed a two-stage semiparametric framework for estimating causal mediation effects with multivariate mediators. The method combines sliced inverse regression for dimension reduction with two single-index models for estimation.

- Research Direction: Multivariate Mediation Analysis via Deep ReLU Networks

Developed a two-stage nonparametric framework for estimating causal mediation effects with multivariate mediators using deep ReLU networks. This project is a long-term extension of our current semiparametric mediation framework, with ongoing work focused on resolving theoretical challenges related to convergence rates, and valid inference.

National Chung Hsing University

Teaching Assistant

Taichung, Taiwan

2021–2023

- Courses: Calculus and Ordinary Differential Equations

PREPRINTS

- [1] L.-S. Zhuang, J.-C. Yu, and Y.-T. Huang, “Nonparametric mediation analysis of non-markov illness-death model”, Manuscript under major revision at *Statistics in Medicine*, 2025.

SCHOLARSHIPS AND AWARDS

- Poster Competition Winner, 2026 ISI–ISM–ISSAS Joint Conference, Academia Sinica 2026
- Prof. Chen Wen-Cheng Statistical Science Scholarship (陳文成統計科學獎學金) 2025

Received for outstanding master’s thesis research in statistical science; awarded annually to only two master’s students.

- Award of Excellence, The 7th Graduate Student Symposium on Statistical Research 2025
- Honorary Member of The Phi Tau Phi Scholastic Honor Society (中華民國斐陶斐榮譽學會榮譽會員) 2022
- Outstanding Teaching Assistant Award – Calculus 2021

CONFERENCE TALKS

- Upcoming: European Meeting of Statisticians, Lugano, Switzerland 2026
Topic: *Nonparametric Mediation Analysis of Non-Markov Illness-Death Model*
- South Taiwan Statistics Conference (南區統計研討會), Taipei, Taiwan 2025
Topic: *Nonparametric Mediation Analysis of Non-Markov Illness-Death Model*
- South Taiwan Statistics Conference, Kaohsiung, Taiwan 2024
Topic: *Causal Mediation of Semi-competing Risks Using Interventional Counterfactual Process*

POSTERS

- ISI-ISM-ISSAS Joint Conference, Institute of Statistical Science, Academia Sinica, Taipei, Taiwan 2026
Topic: *Nonparametric Mediation Analysis of Non-Markov Illness-Death Model*
- The 7th Graduate Student Symposium on Statistical Research, Taichung, Taiwan 2025
Topic: *Nonparametric Mediation Analysis of Non-Markov Illness-Death Model*
- Poster Competition, Institute of Statistical Science, Academia Sinica, Taipei, Taiwan 2025
Topic: *Causal Mediation of Semi-competing Risks Using Interventional Counterfactual Process*

SELECTED PROJECTS

- A General Paired t -Test under Non-i.i.d. Assumptions 2024
Applied the Lyapunov Central Limit Theorem and k -nearest neighbors to extend the classical paired t -test framework.
- Fourier-Based Stock Price Modeling 2024
Applied the Fourier transform to denoise log returns in stock price data, improving prediction accuracy in a time series forecasting project.

TECHNICAL SKILLS

I have experience using R, Python, and Matlab for data analysis, model development, and statistical computing.